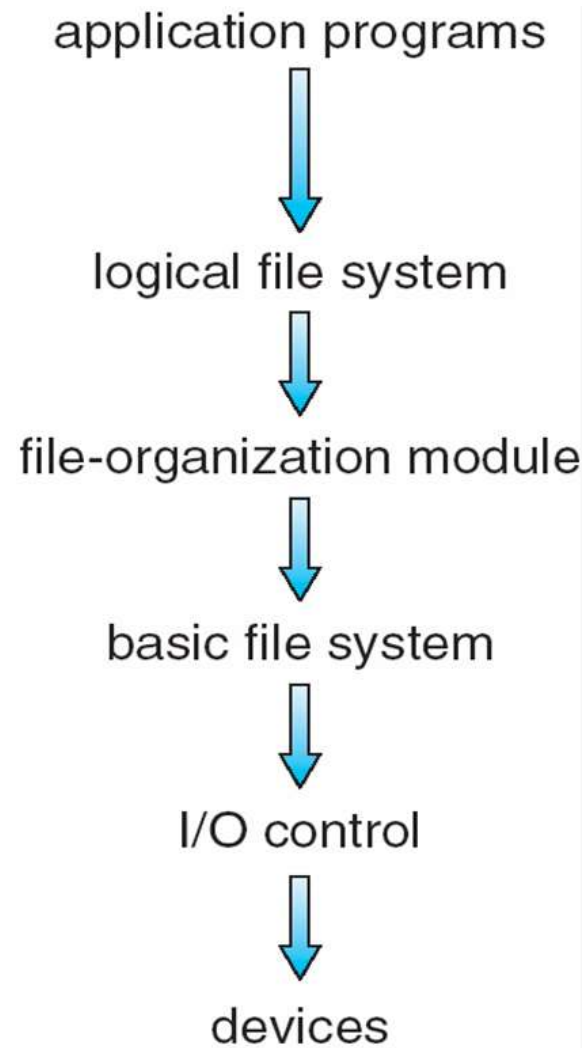


Layered File System



Device Drivers

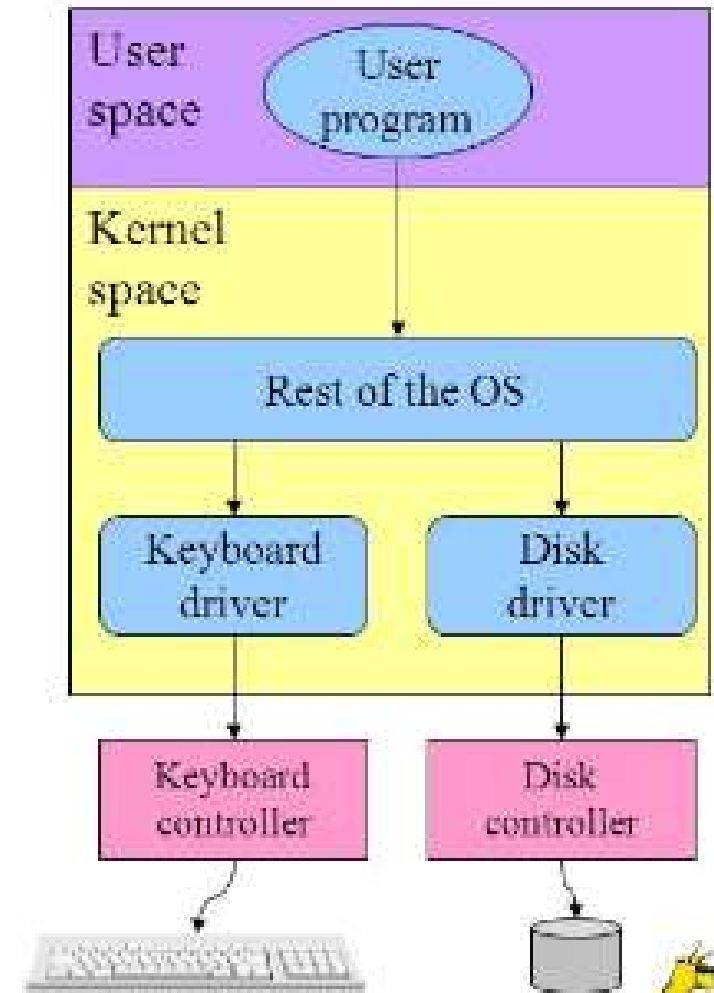
- **Device drivers** manage I/O devices at the I/O control layer
- **Device driver** controls the physical device
- The I/O control consists of
 - device drivers and
 - interrupt handlers
 - to transfer information between the main memory and the disk system.

Device Drivers

- A device driver
 - can be thought of as a translator.
 - **Its input consists of high level commands such as “retrieve block 123”**
 - **Given commands like “read drive1, cylinder 72, track 2, sector 10, into memory location 1060”**
 - **outputs low-level hardware specific commands to hardware controller**

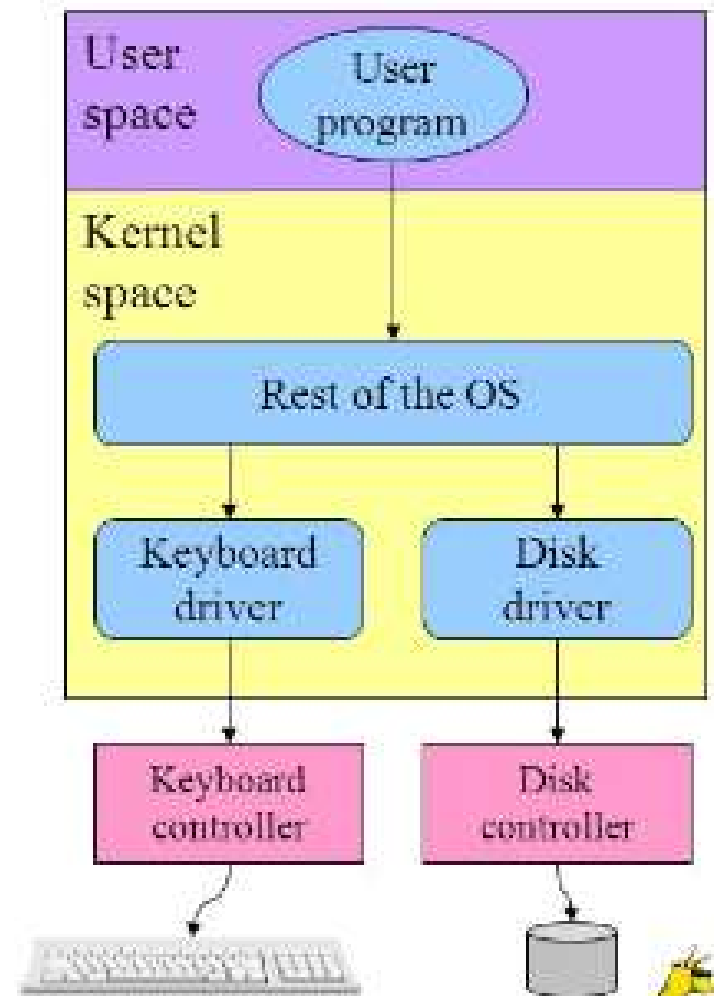
Device Drivers

- Its output consists of
 - **low level hardware specific instructions**
 - **that are used by the hardware controller**
 - which interfaces I/O device to the rest of the system.
- The device driver
 - **writes specific bit patterns**
 - **to special locations in the I/o controller's memory**
 - to tell the controller on **which device location to act and**
 - **what actions to take.**



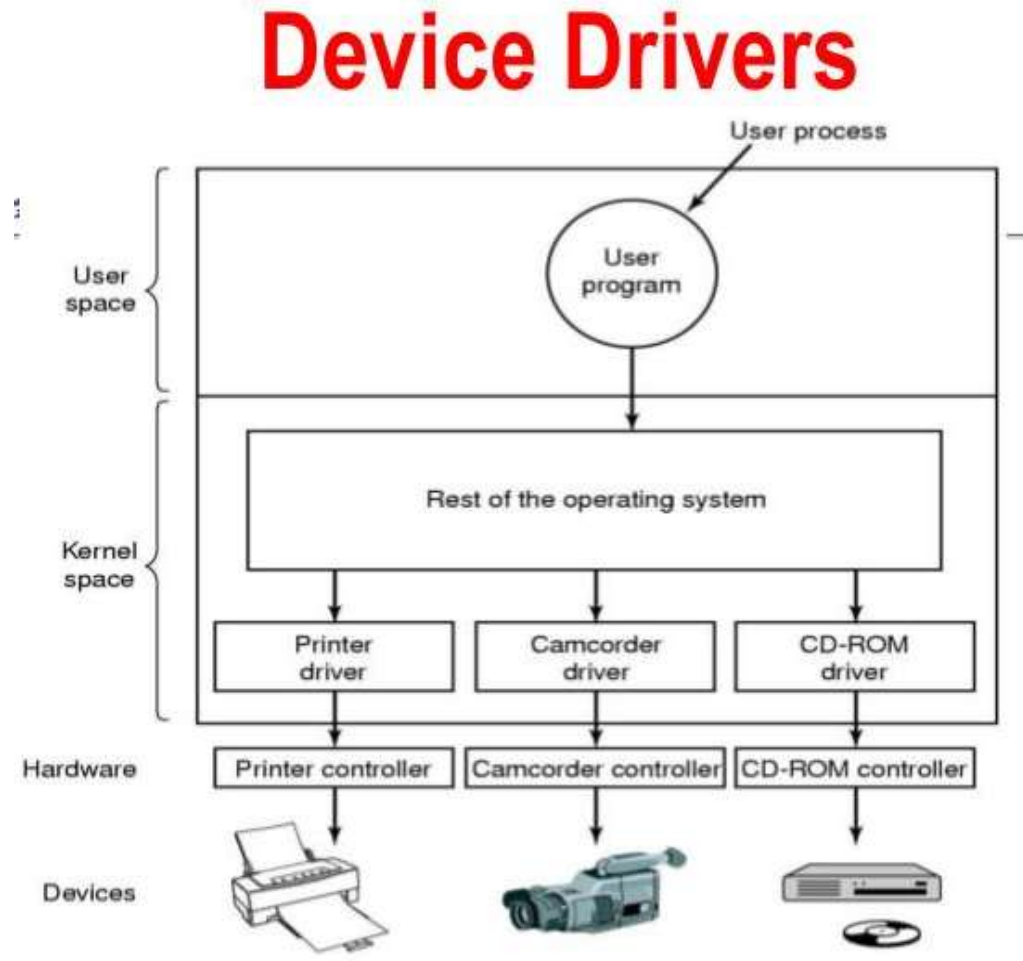
Device Drivers

- Device drivers are
 - **software modules that can be plugged into an OS to handle a particular device.**
- Operating System takes help from
 - device drivers to handle all I/O devices.



Device Controllers

- The Device Controller works like
 - an interface between a device and a device driver.
- There is always a device controller and a device driver
 - for each device to communicate with the Operating Systems.



Device Controllers

- I/O units (Keyboard, mouse, printer, etc.) typically consist of
 - **a mechanical component and**
 - **an electronic component**
 - **where electronic component is called the device controller.**

Courtesy:https://www.tutorialspoint.com/operating_system/os_io_hardware.htm

Device Controllers

- **A Electrical component/device controller controls the device**
 - May be able to handle multiple devices.
 - May be more than one controller per mechanical component. (Eg- Hard drive)
- **Controller's Tasks-**
 - Convert serial bit stream to block of bytes,
 - **Perform error correction as necessary**
 - **Make device available to main memory**

Courtesy:https://www.tutorialspoint.com/operating_system/os_io_hardware.htm